

Co-Designing MTIA's Communication Stack

Kirtesh Patil
Tech Lead, AI Networking



Agenda

- 01 Why communication is the critical path
- 02 MTIA: On package communication & networking
- 03 Message Offloading
- 04 The latency path
- 05 Ethernet + network fungibility

Workload Evolution

- Ranking & Recommendation (R&R)
- Generative AI
- Agentic use-cases

Model & Data Scaling

- R&R models tend to have large embedding tables and training time is dominated by collectives like all-to-all
- Generative AI models have similar challenges with respect to weights, activations and tokens.

MTIA



Messaging

16x engines for messaging, namely, ME (Message Engines).

Reduction engines, namely, NMC (Near Memory Compute) for collectives like AllReduce



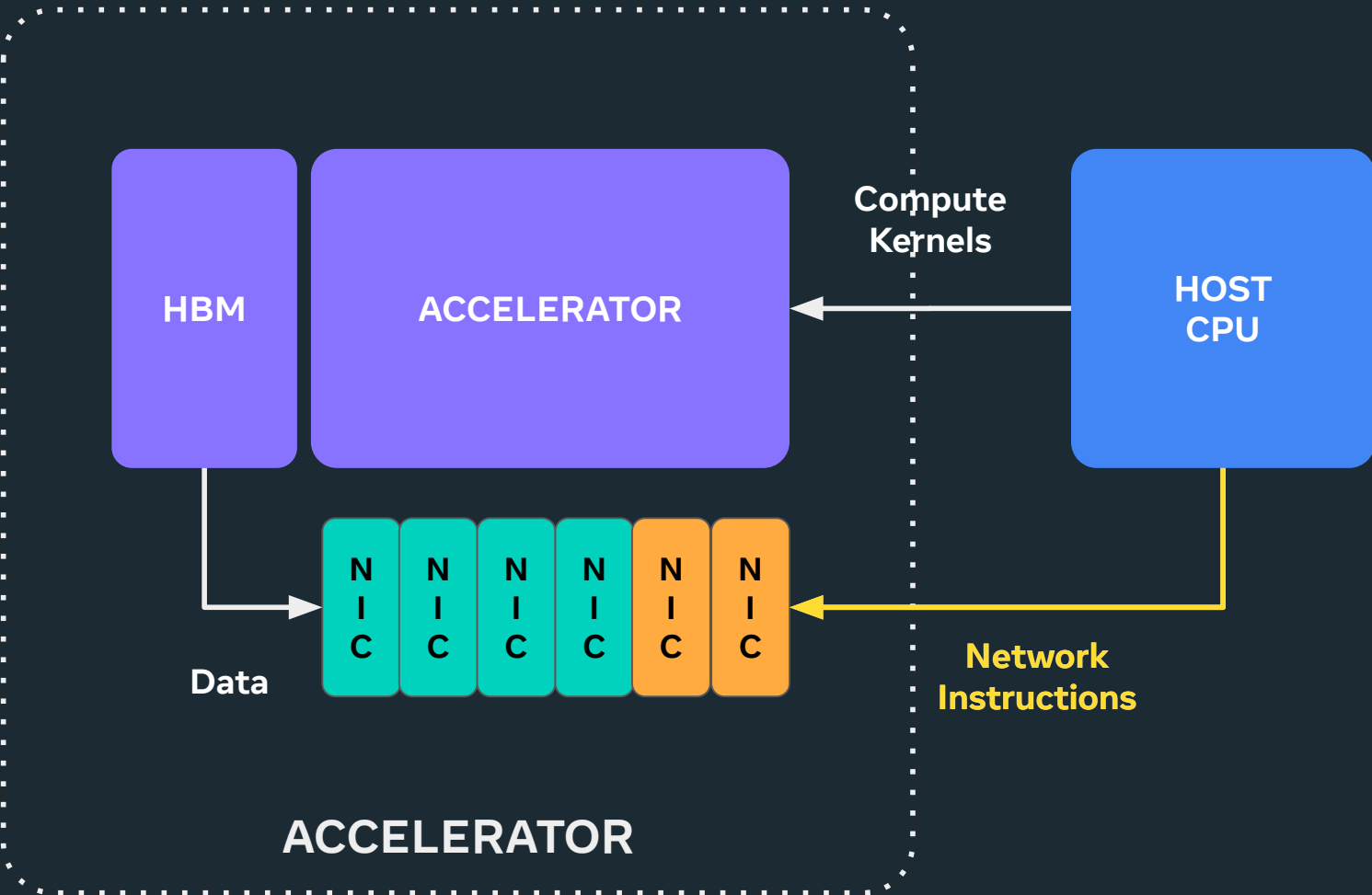
Network

12x Integrated RDMA NICs for I/O

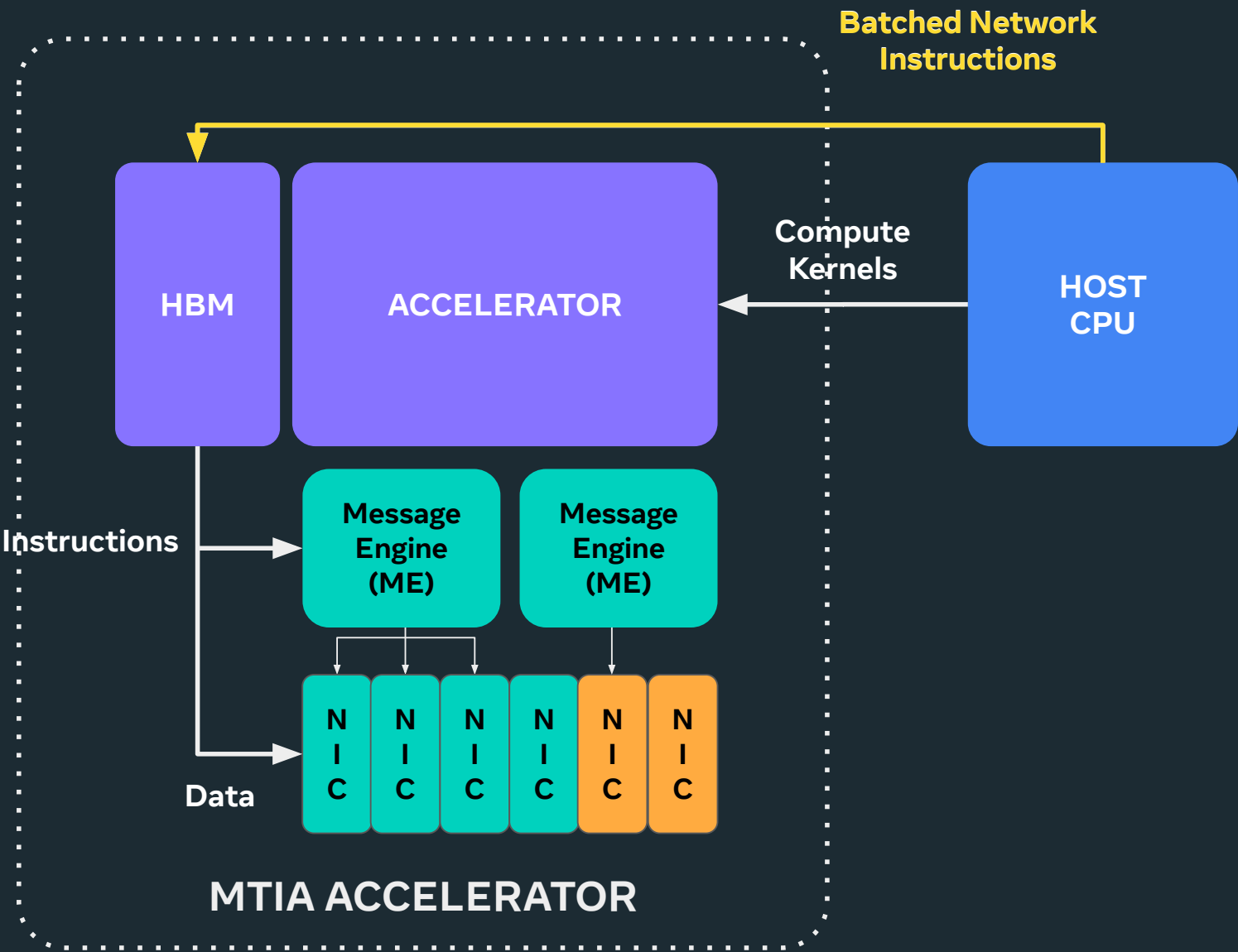
MEs have ability to issue work-request and get completions to & from the the NICs

03 Message Offloading

HPC/AI SYSTEM



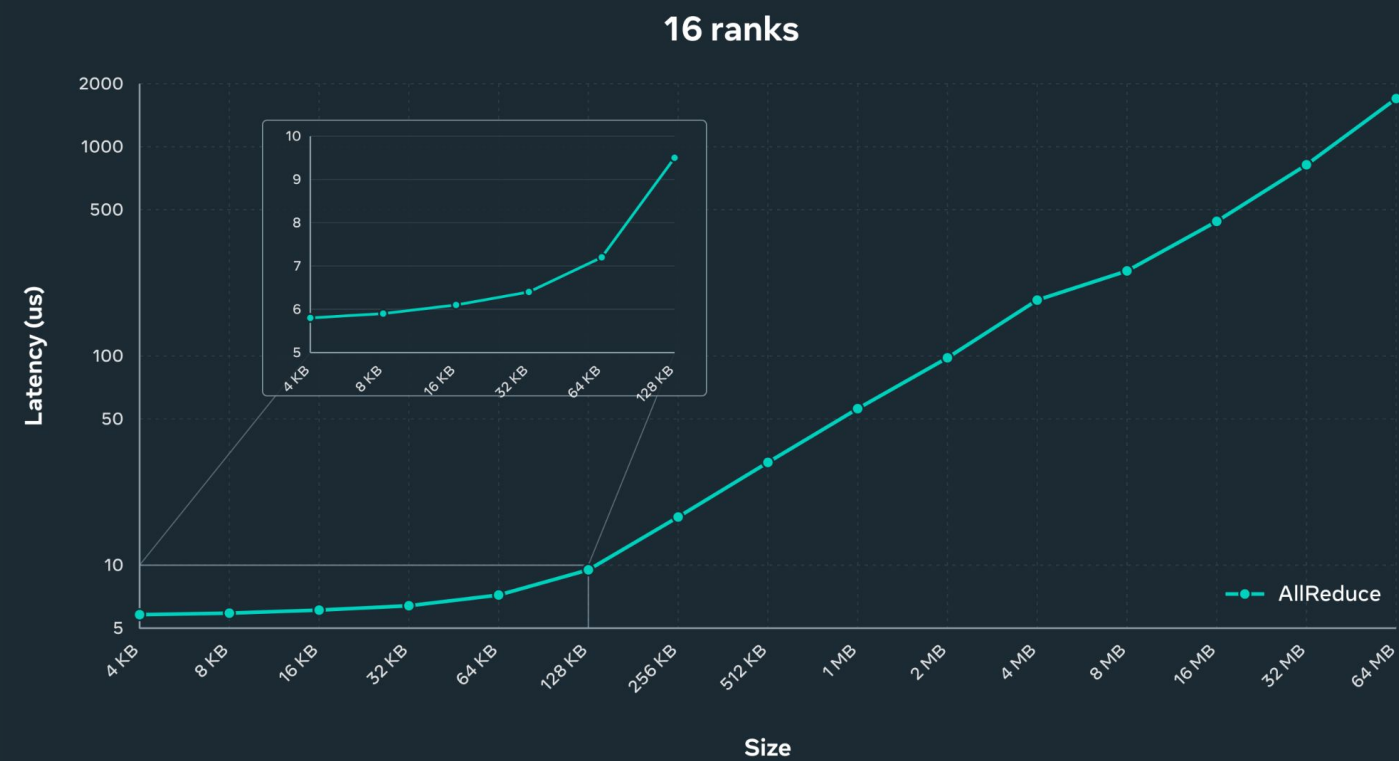
MTIA 300 SYSTEM



Offloading:

- Improved communication efficiency
- Requires more work upfront which is acceptable for largest collective.

Latency sensitive paths require direct work-request injection from the PEs.



Scale-up & scale-out fungibility

- **Using ethernet for both is necessary and sufficient**
- All NICs can be used for scale-up or scale-out
- Can shift to an all-scale-up configuration
- Different cabling and network setup
- Can also use scale-out NICs for scale-up comms

Default configuration: 10 scale-up NICs & 2 scale-out NICs

- Strikes balance for training-focused workloads

